Candidate Seat no............

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

**Six Semester of B.Tech. (IT) Examination**
**May 2022**

## IT378 WIRELESS COMMUNICATION & MOBILE COMPUTING

**Date: 14/05/2022, Saturday** **Time: 10:00 a.m. to 01:00 p.m.** **Maximum Marks: 70**

*Instructions:*

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

## SECTION – I

**Q - 1** **Do as directed.** **[07]**

(a). In CDMA, there are two sender A & B, sender A send data $A_d = 1$, Key $A_k = 010000$ sender B send data $B_d = 0$, Key $B_k = 011100$ (assign "0" = -1 & "1" = +1). Combine signal ( $A_s + B_s$) will send to receiver, but receiver only want to receive from sender B. So, find out Combine signal value & at receiver value from Sender B. [02]

**(b)** Following are the hiperlan layers. [01]

a. Network layer, data link layer, physical layer
b. Logical Link Layer, MAC layer, Application layer
c. Network layer, MAC layer, Channel access control layer
d. MAC layer, Channel access control layer, physical layer

**(c)** Which information is not stored in HLR? [01]

a. list of IMEIs of all valid terminals
b. Identification information
c. Storage of SMS Service Center (SC) number
d. International Mobile Subscriber Identity (IMSI)

**(d)** Which one is correct UMTS hierarchical cell structure in ascending order? [01]

a. World, Macro , Micro, Pico
b. Macro , World , Micro, Pico
c. Pico, Micro , Macro, World
d. Pico , Macro , Micro, World

**(e)** ______ is the entire set of cells served by one network operator and is defined as the area in which an operator offers radio coverage and access to its network. [01]

a. GPRS
b. GSM
c. PLMN
d. Satellite

(f) _________ converts GPRS Packets Data Protocol (PDP) into the corresponding structure. [01]

a. BSC
b. SGSN
c. GMSC
d. GGSN

**Q – 2 (a)** Explain GSM operational architecture system with proper diagram. **[06]**

**OR**

**Q – 2 (a)** Explain UMTS architecture with proper diagram. **[06]**

**Q – 2 (b)** How authentication and security are provided in GSM? **[04]**

**OR**

**Q – 2 (b)** Explain Modulation and demodulation with block diagram. **[04]**

**Q – 2 (c)** Why frequency reuse is required? Explain three cell structure, three cell cluster with three sector antennas. **[04]**

**OR**

**Q – 2 (c)** What is Handover in GSM? Explain four types of handovers with diagram. **[04]**

**Q - 3 (a)** **Explain the following terms.** **[04]**

1. Paging
2. Refraction
3. Mobile Station
4. IMSI

**Q - 3 (b)** GSM use which multiplexing methods? Explain with diagram. **[05]**

**OR**

**Q - 3 (b)** Write a short note on GPRS. **[05]**

**Q - 3 (c)** Explain Satellite system classes. **[05]**

## SECTION – II

**Q - 4** **Do as directed** **[07]**

(a) ______ is the system (node) that can change the point of connection to the network without changing its IP address. [01]

a. Foreign Agent (FA)
b. Correspondent Node (CN)
c. Home Agent (HA)
d. None of above

(b) The ______ approach has the same goals as I-TCP and snooping TCP. [01]

a. I-TCP
b. TCP
c. Snooping TCP
d. M-TCP

(c) _________ and ________ addresses are required to form a piconet. [02]

(d) Explain Mobile IP registration request and reply using Sequence diagram. [03]

Q - 5 (a) Discuss Indirect TCP (I-TCP) along with its advantages and disadvantages. How snooping TCP maintain end to end semantics with suitable sketch and sequence diagram. [06]

**OR**

Q - 5 (a) Compare Traditional TCP with I-TCP [06]

Q - 5 (b) Explain Piconet & Scatternet. What are the different state of devices in piconet? [05]

Q - 5 (c) Compare two wireless technology WiFi vs WiMax. [05]

Q - 6 (a) Explain Dynamic Host Configuration Protocol mechanism. [05]

**OR**

Q - 6 (a) Explain DSDV protocol with example. What are the Problems of traditional routing algorithms? [05]

Q - 6 (b) Explain architecture of Mobile IP with figure. [07]

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